

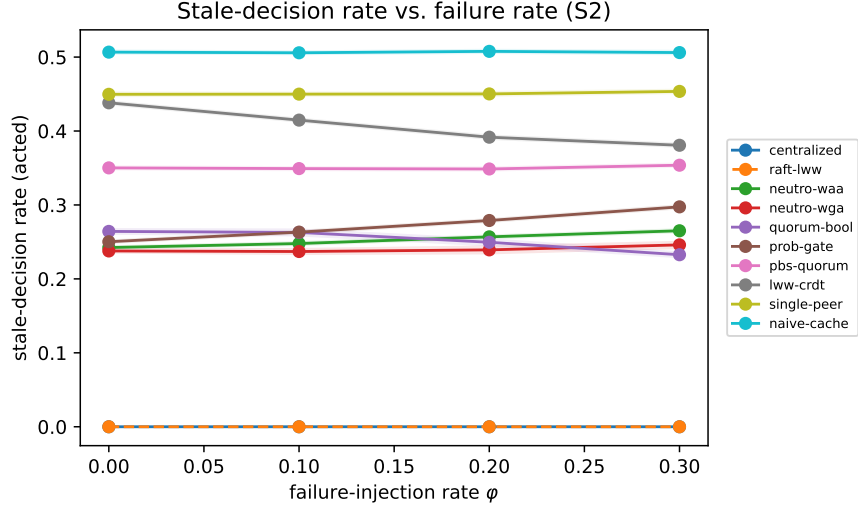


Figure S1: Stale-decision rate vs. failure-injection rate (S2). Bands are 5000-resample bootstrap 95% CIs.

## Supplementary Information

A decentralized neutrosophic decision layer for global-state freshness in microservices architectures

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## Supplementary Figures

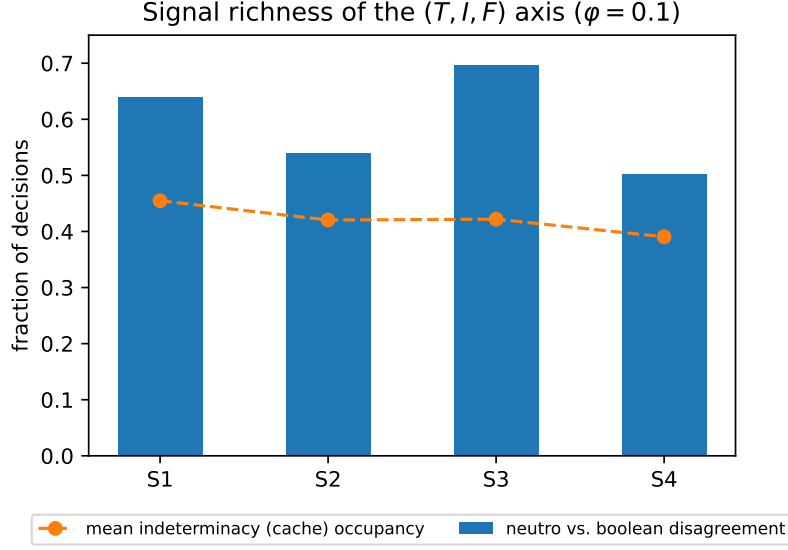


Figure S2: RQ1: the indeterminacy axis is well populated and the  $(T, I, F)$  decision diverges from a boolean quorum on a substantial fraction of decisions.

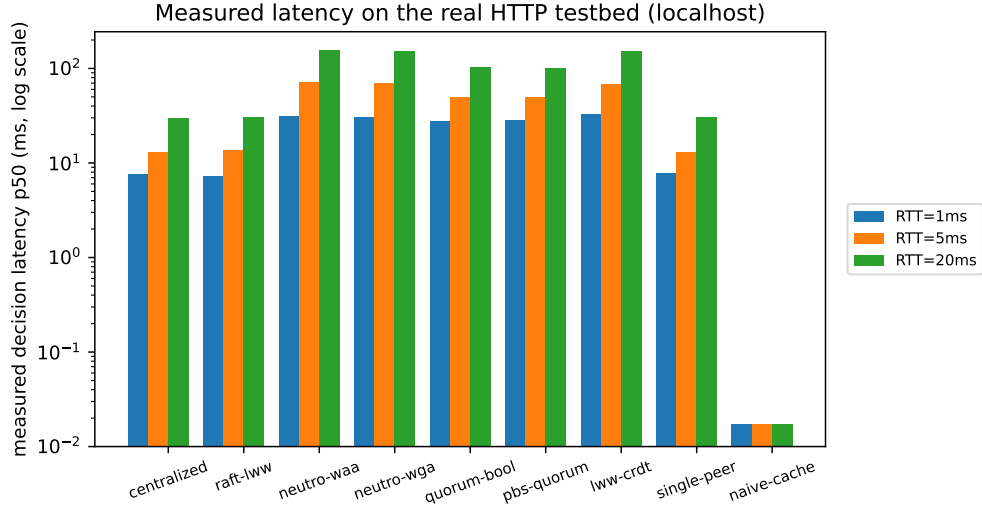


Figure S3: Measured median decision latency on the real localhost HTTP testbed: local read, single round-trip, and  $R$ -peer fan-out tiers. Log scale: naive-cache is a local read ( $\approx 0.02$  ms), dwarfed by the network tiers but fully measured ( $n = 200$  per cell).

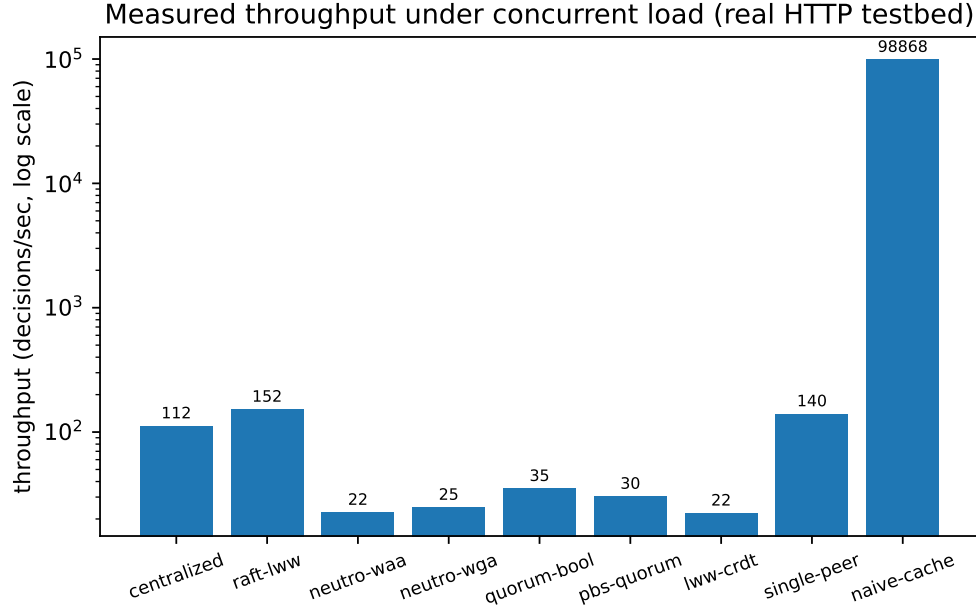


Figure S4: Measured throughput (decisions/sec) under concurrent closed-loop load on the real HTTP testbed.

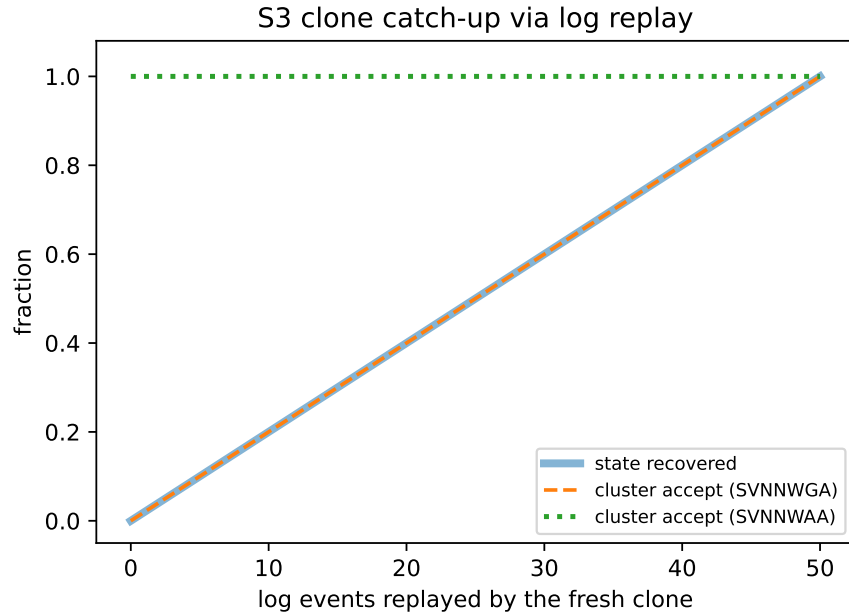


Figure S5: S3: a fresh clone recovers state by replaying the log; SVNNWGA acceptance trails until catch-up while SVNNWAA stays available throughout. State-recovery and SVNNWGA acceptance coincide (dashed line over the thick one).

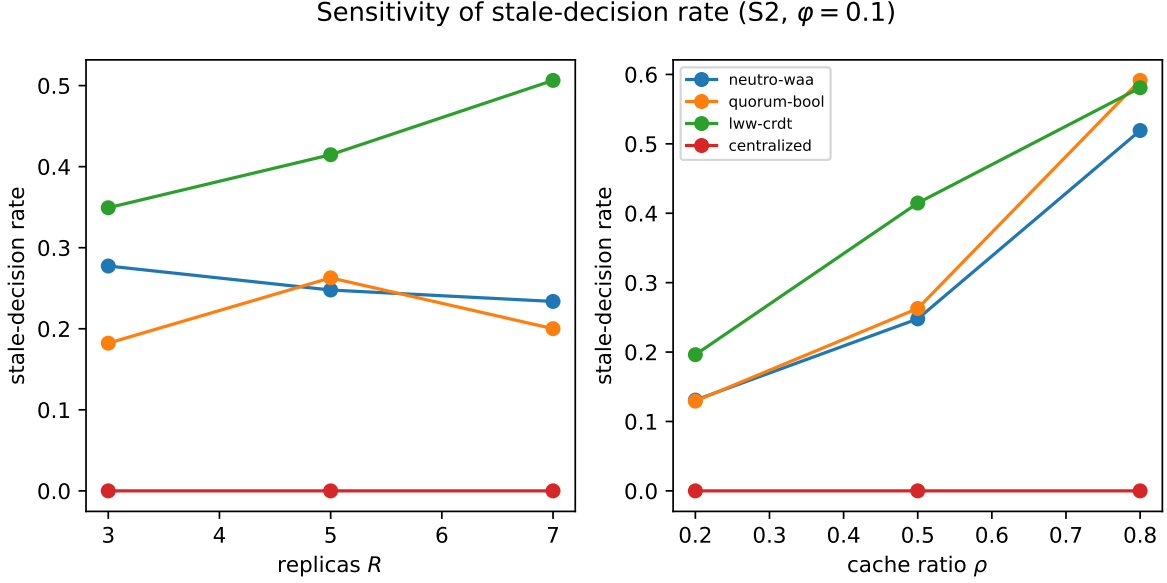


Figure S6: Sensitivity of the stale-decision rate to replica count  $R$  and cache ratio  $\rho$  ( $S2, \varphi = 0.1$ ).

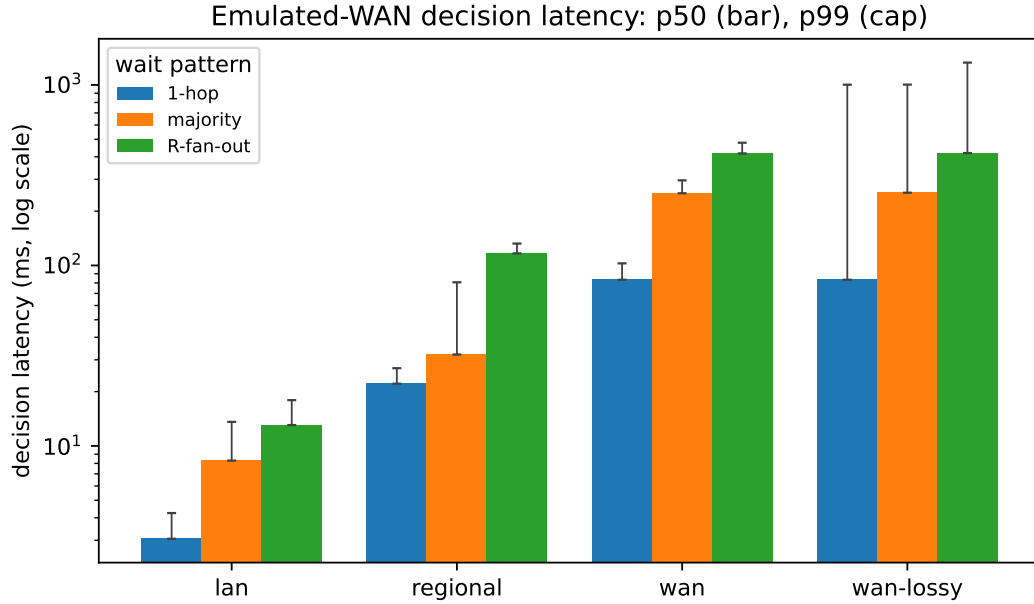


Figure S7: Decision latency on a single-host Docker deployment under *emulated* WAN (Toxiproxy per-link latency, jitter, and loss): p50 (bar) and p99 (cap), log scale. The wait patterns (single-hop, majority,  $R$ -peer fan-out) separate cleanly and the gap grows with link latency; under 2% loss the tails inflate to the 1s decision deadline, most for the fan-out (it waits for the slowest of  $R$  peers). Real HTTP over proxied sockets; not geo-distributed.

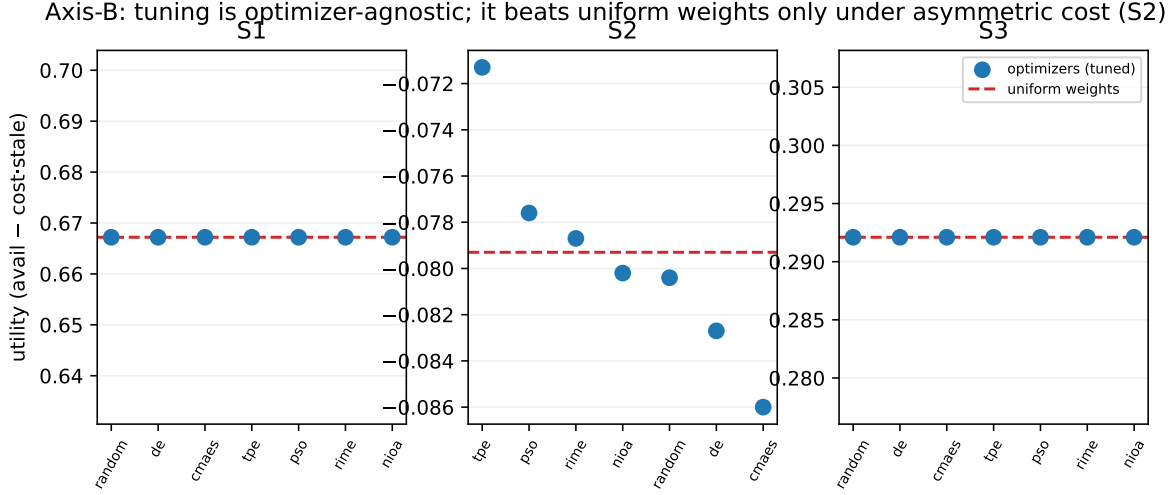


Figure S8: Axis B: equal-budget metaheuristic tuning of  $[\tau, \text{weights}]$  (seven optimizers, 120 evaluations each), tuned utility per optimizer (points) versus the uniform-weights baseline (dashed). At equal budget the optimizers converge—identically on the symmetric S1/S3, within  $\approx 2\%$  on the asymmetric S2—and tuning beats uniform weights only on S2 (+10.1%); on S1/S3 uniform is already optimal. No-Free-Lunch: the gain is not an artefact of the optimizer.

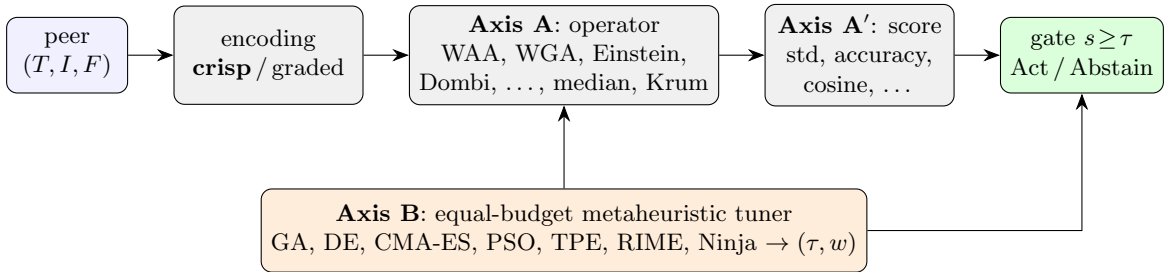


Figure S9: The two-axis audit pipeline. Peer views are encoded (crisp anchor or graded), aggregated by an operator-panel member (Axis A), deneutrosophized by a score-panel member (Axis A'), and gated at the calibrated threshold; an equal-budget metaheuristic panel (Axis B) tunes the threshold and peer weights. One axis is varied at a time, holding the others at the submitted anchor.

Table S1: Graded-encoding anchors (the GRADE map of Algorithm 2). A peer’s observable kind and version-lag  $\ell$  relative to the most up-to-date reachable peer (freshness  $\varphi = 1/(1 + \ell/\sigma)$ , softness  $\sigma = 2$ ) map to a single-valued neutrosophic triple  $(T, I, F)$ :  $T$  is confidence the held value is the committed-current truth,  $I$  is unconfirmedness,  $F$  is confidence the value is stale, superseded, or absent. The anchors are a documented, tunable design choice (calibration override `grade_encoding`); the crisp corners  $(1, 0, 0)/(0, 1, 0)/(0, 0, 1)$  remain the conservative baseline.

| Peer kind (observable)                           | $T$           | $I$  | $F$  |
|--------------------------------------------------|---------------|------|------|
| Persisted, fresh ( $\ell = 0$ )                  | 1.00          | 0.10 | 0.00 |
| Persisted, lagging ( $\ell \rightarrow \infty$ ) | 0.50          | 0.10 | 0.40 |
| Clean cache, fresh                               | 0.45          | 0.55 | 0.10 |
| Clean cache, lagging                             | 0.00          | 0.65 | 0.35 |
| Suspected dirty write                            | $0.15\varphi$ | 0.45 | 0.45 |
| Absent / unreachable                             | 0.05          | 0.10 | 0.85 |

## Supplementary Tables

Table S2: Headline results for S2 at  $\varphi = 0.1$  (30 reps/cell, 3 seeds; stale-rate with 5000-resample bootstrap 95% CI). All cells **validated**.

| System            | Partition | Stale-rate [95% CI]  | Availability | Msgs/dec |
|-------------------|-----------|----------------------|--------------|----------|
| Centralized       | none      | 0.000 [0.000, 0.000] | 0.900        | 1        |
| Raft-LWW          | none      | 0.000 [0.000, 0.000] | 0.900        | 1        |
| <b>Neutro-WAA</b> | none      | 0.248 [0.244, 0.251] | 0.942        | 5        |
| <b>Neutro-WGA</b> | none      | 0.237 [0.232, 0.242] | 0.427        | 5        |
| Quorum-bool       | none      | 0.263 [0.257, 0.268] | 0.402        | 5        |
| PBS-quorum        | none      | 0.349 [0.346, 0.353] | 1.000        | 3        |
| LWW-CRDT          | none      | 0.415 [0.411, 0.418] | 1.000        | 5        |
| Single-peer       | none      | 0.450 [0.446, 0.453] | 1.000        | 1        |
| Naive-cache       | none      | 0.506 [0.503, 0.509] | 1.000        | 0        |
| Centralized       | transient | 0.000 [0.000, 0.000] | 0.453        | 1        |
| Raft-LWW          | transient | 0.000 [0.000, 0.000] | 0.453        | 1        |
| <b>Neutro-WAA</b> | transient | 0.263 [0.260, 0.266] | 0.880        | 5        |
| <b>Neutro-WGA</b> | transient | 0.240 [0.235, 0.245] | 0.332        | 5        |
| Quorum-bool       | transient | 0.223 [0.218, 0.227] | 0.410        | 5        |
| PBS-quorum        | transient | 0.348 [0.345, 0.351] | 0.996        | 3        |
| LWW-CRDT          | transient | 0.380 [0.377, 0.384] | 0.996        | 5        |
| Single-peer       | transient | 0.450 [0.446, 0.453] | 0.996        | 1        |
| Naive-cache       | transient | 0.506 [0.503, 0.510] | 1.000        | 0        |

Table S3: Headline results for S1 at  $\varphi = 0.1$  (30 reps/cell, 3 seeds; stale-rate with 5000-resample bootstrap 95% CI). All cells **validated**.

| System            | Partition | Stale-rate [95% CI]  | Availability | Msgs/dec |
|-------------------|-----------|----------------------|--------------|----------|
| Centralized       | none      | 0.000 [0.000, 0.000] | 0.901        | 1        |
| Raft-LWW          | none      | 0.000 [0.000, 0.000] | 0.901        | 1        |
| <b>Neutro-WAA</b> | none      | 0.137 [0.134, 0.139] | 1.000        | 5        |
| <b>Neutro-WGA</b> | none      | 0.106 [0.103, 0.109] | 0.427        | 5        |
| Quorum-bool       | none      | 0.106 [0.102, 0.109] | 0.361        | 5        |
| PBS-quorum        | none      | 0.272 [0.269, 0.275] | 1.000        | 3        |
| LWW-CRDT          | none      | 0.343 [0.340, 0.347] | 1.000        | 5        |
| Single-peer       | none      | 0.327 [0.324, 0.331] | 1.000        | 1        |
| Naive-cache       | none      | 0.386 [0.382, 0.390] | 1.000        | 0        |
| Centralized       | transient | 0.000 [0.000, 0.000] | 0.452        | 1        |
| Raft-LWW          | transient | 0.000 [0.000, 0.000] | 0.452        | 1        |
| <b>Neutro-WAA</b> | transient | 0.171 [0.168, 0.174] | 0.997        | 5        |
| <b>Neutro-WGA</b> | transient | 0.109 [0.105, 0.114] | 0.318        | 5        |
| Quorum-bool       | transient | 0.089 [0.085, 0.092] | 0.370        | 5        |
| PBS-quorum        | transient | 0.269 [0.266, 0.272] | 0.997        | 3        |
| LWW-CRDT          | transient | 0.306 [0.303, 0.309] | 0.997        | 5        |
| Single-peer       | transient | 0.327 [0.324, 0.331] | 0.997        | 1        |
| Naive-cache       | transient | 0.388 [0.384, 0.391] | 1.000        | 0        |